

Problem 2.1. Prove the “homogeneous Nullstellensatz,” which says if $\mathfrak{a} \subseteq S$ is a homogeneous ideal, and if $f \in S$ is a homogeneous polynomial with $\deg f > 0$, such that $f(P) = 0$ for all $P \in Z(\mathfrak{a})$ in $\mathbf{P}^n$, then $f^q \in \mathfrak{a}$ for some $q > 0$.

[Hint: Interpret the problem in terms of the affine $(n + 1)$ -space whose affine coordinate ring is S , and use the usual Nullstellensatz, (1.3A).]

Problem 2.2. For a homogeneous ideal $\mathfrak{a} \subseteq S$, show that the following conditions are equivalent:

(i) $Z(\mathfrak{a}) = \emptyset$ (the empty set);

(ii) $\sqrt{\mathfrak{a}}$ is either S or the ideal

$$S_+ = \bigoplus_{d>0} S_d;$$

(iii) $\mathfrak{a} \supseteq S_d$ for some $d > 0$.

Problem 2.3.

(a) If $T_1 \subseteq T_2$ are subsets of S^h , then

$$Z(T_1) \supseteq Z(T_2).$$

(b) If $Y_1 \subseteq Y_2$ are subsets of $\mathbf{P}^n$, then

$$I(Y_1) \supseteq I(Y_2).$$

(c) For any two subsets Y_1, Y_2 of $\mathbf{P}^n$,

$$I(Y_1 \cup Y_2) = I(Y_1) \cap I(Y_2).$$

(d) If $\mathfrak{a} \subseteq S$ is a homogeneous ideal with $Z(\mathfrak{a}) \neq \emptyset$, then

$$I(Z(\mathfrak{a})) = \sqrt{\mathfrak{a}}.$$

(e) For any subset $Y \subseteq \mathbf{P}^n$,

$$Z(I(Y)) = \overline{Y}.$$

Problem 2.4.

- (a) There is a one-to-one inclusion-reversing correspondence between algebraic sets in $\mathbf{P}^n$, and homogeneous radical ideals of S not equal to S_+ , given by

$$Y \longmapsto I(Y) \quad \text{and} \quad \mathfrak{a} \longmapsto Z(\mathfrak{a}).$$

Note: Since S_+ does not occur in this correspondence, it is sometimes called the irrelevant maximal ideal of S .

- (b) An algebraic set $Y \subseteq \mathbf{P}^n$ is irreducible if and only if $I(Y)$ is a prime ideal.
 (c) Show that $\mathbf{P}^n$ itself is irreducible.

Problem 2.5.

- (a) $\mathbf{P}^n$ is a noetherian topological space.
 (b) Every algebraic set in $\mathbf{P}^n$ can be written uniquely as a finite union of irreducible algebraic sets, no one containing another. These are called its *irreducible components*.

Problem 2.6. If Y is a projective variety with homogeneous coordinate ring $S(Y)$, show that

$$\dim S(Y) = \dim Y + 1.$$

[Hint: Let $\varphi_i : U_i \rightarrow \mathbb{A}^n$ be the homeomorphism of (2.2), let Y_i be the affine variety $\varphi_i(Y \cap U_i)$, and let $A(Y_i)$ be its affine coordinate ring. Show that $A(Y_i)$ can be identified with the subring of elements of degree 0 of the localized ring $S(Y)_{x_i}$. Then show that

$$S(Y)_{x_i} \cong A(Y_i)[x_i, x_i^{-1}].$$

Now use (1.7), (1.8A), and (Ex. 1.10), and look at transcendence degrees. Conclude also that $\dim Y = \dim Y_i$ whenever Y_i is nonempty.]